

Alumni Networking Portal

with Job Posting and Communication Features
Avanthi Institute of Engineering and Technology, Makavarapalem.

Presented By:

- M. Bhargavi (22811A05A1)
- K. Murali Karthik (22811A0583)
- P. Maha Lakshmi (22811A05B7)
- K. Chinni Krishna (22811A0576)
- M. Pranay Kumar (23815A0519)
- L. Rama Krishna (23815A0515)

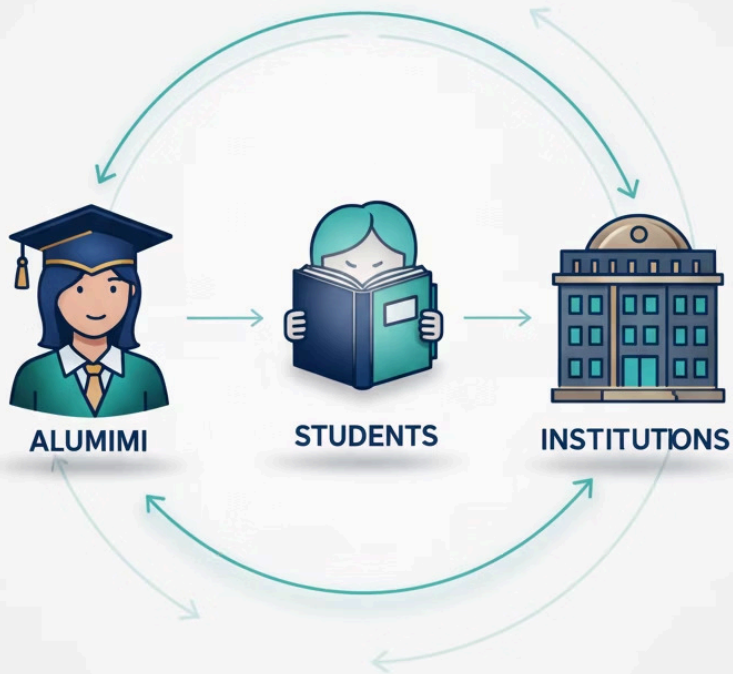
Project Guide:

Miss. Dhana Lakshmi, M.Tech
Assistant Professor
Department of CSE



Presentation Overview

 Introduction	 Project Abstract
 Existing System	 Proposed System
 Literature Review	 System Architecture
 Software Requirement Specification	 Modules
 Data Flow Diagram	 UML Diagrams
 Conclusion	 References



PROJECT OVERVIEW

What is the Project?

A centralized and secure alumni networking platform that connects verified alumni, students, and institutions for job opportunities, mentorship and professional communication — powered by AI-driven insights.

- **Alumni Share Opportunities:** Post job openings, mentorship offers, and announcements
- **Students Discover & Connect:** Explore opportunities, apply for jobs, and communicate with alumni
- **Institutions Gain Insights:** Access verified alumni data and career analytics

Project Context & Scope

Modern institutions struggle with effectively engaging growing student and alumni populations, tracking career outcomes, and understanding long-term professional impact. This project provides a centralized platform to overcome these challenges, fostering sustained alumni interaction and seamless academic-industry continuity.

Project Scope Includes

- Academic and career engagement
- Centralized communication
- Career outcome visibility
- Alumni journey insights

Stakeholders Covered

- Students
- Alumni
- Faculty & Administrators



Existing Solutions & Their Limitations

Current tools used by students and institutions



LinkedIn : Not institution-specific, lacks verification



WhatsApp Groups: Informal, unstructured, no verification

Why They Fail

- Not institution-specific or verified
- Information scattered across platforms
- No structured job tracking
- Manual and time-consuming processes
- No analytics or long-term engagement



Email-based systems: Unstructured and difficult to track



Facebook Groups: Open platform, low authenticity, poor control

Gaps in Alumni Engagement

- Inefficient Job Matching
- Limited Institutional Insight
- Suboptimal Communication

Proposed Solution

The Alumni Networking Portal is a centralized, web-based system designed to connect verified alumni, students, and administrators for job sharing, mentorship, and communication, ensuring secure access, efficient data management, and institutional oversight.



Alumni



Students



Administrator



Central Portal

Core Functional Features



Alumni Registration & Profile Management
Secure registration with robust profile verification.



Alumni Directory with Search
Searchable database to discover alumni by skills, location, and industry



Job & Internship System
Alumni-driven opportunities with structured applications.



Secure Communication
Trusted messaging for alumni-student mentorship.



Admin Management
User approval, access control, and platform governance.



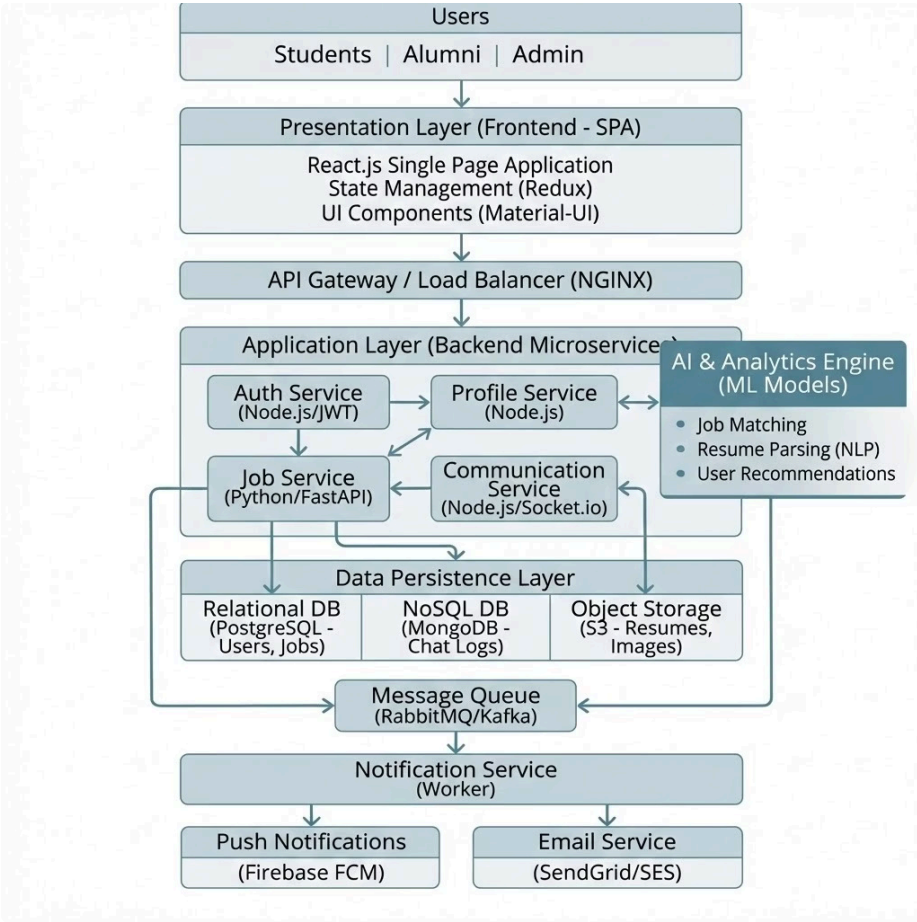
Event Announcements & Notifications
Real-time updates on networking events and opportunities

Literature Review

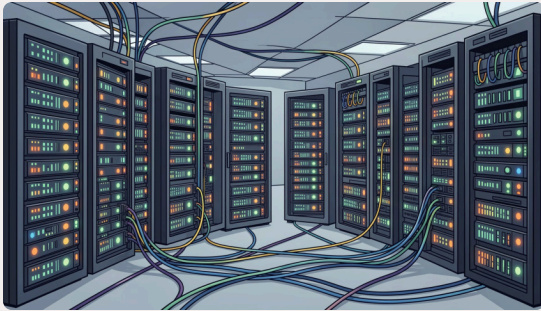
A structured comparison of current platforms and research, highlighting deficiencies our project aims to address.

Platform / Research	Key Features	Identified Gaps
LinkedIn	Professional networking, job search, profile visibility	Not institution-specific, no alumni verification system
AlmaConnect	Alumni engagement platform, job postings, event management	Subscription-based model, limited AI integration
IEEE Alumni Management System	Centralized alumni database, structured communication	No AI-based personalization, limited job recommendation features
AI-Based Job Recommendation Systems	Resume-based job matching, machine learning models	Not integrated with alumni networking platforms

System Architecture



Software Requirement Specification (SRS)



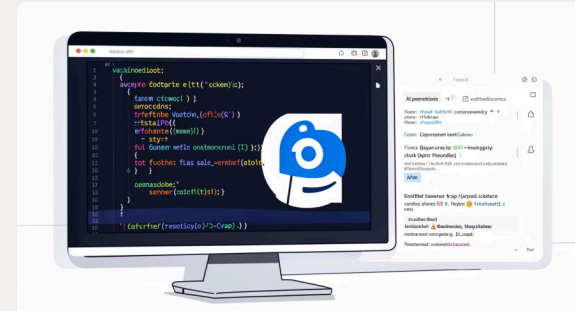
Hardware Requirements

Server :

- Intel i3 or above
- 8GB RAM minimum
- 256GB SSD

Client:

- 4GB RAM
- Modern browser (Chrome/Edge/Firefox)
- Internet connection



Software Requirements

- Frontend: HTML, CSS, JavaScript, React
- Backend: Flask / Node.js
- Database: MySQL / MongoDB
- Tools: VS Code, GitHub

System Modules

User Management & Authentication Module

Secure, role-based login system categorizing users into Students, Alumni, and Administrators.

AI-Powered Job Board Module

Alumni can post jobs while the system's background Machine Learning service extracts data from uploaded student PDF resumes to generate semantic embeddings and calculate job match scores.

Directory & Search Module

Provides filterable talent grids and alumni directories, allowing users to search by department, location, and skills.

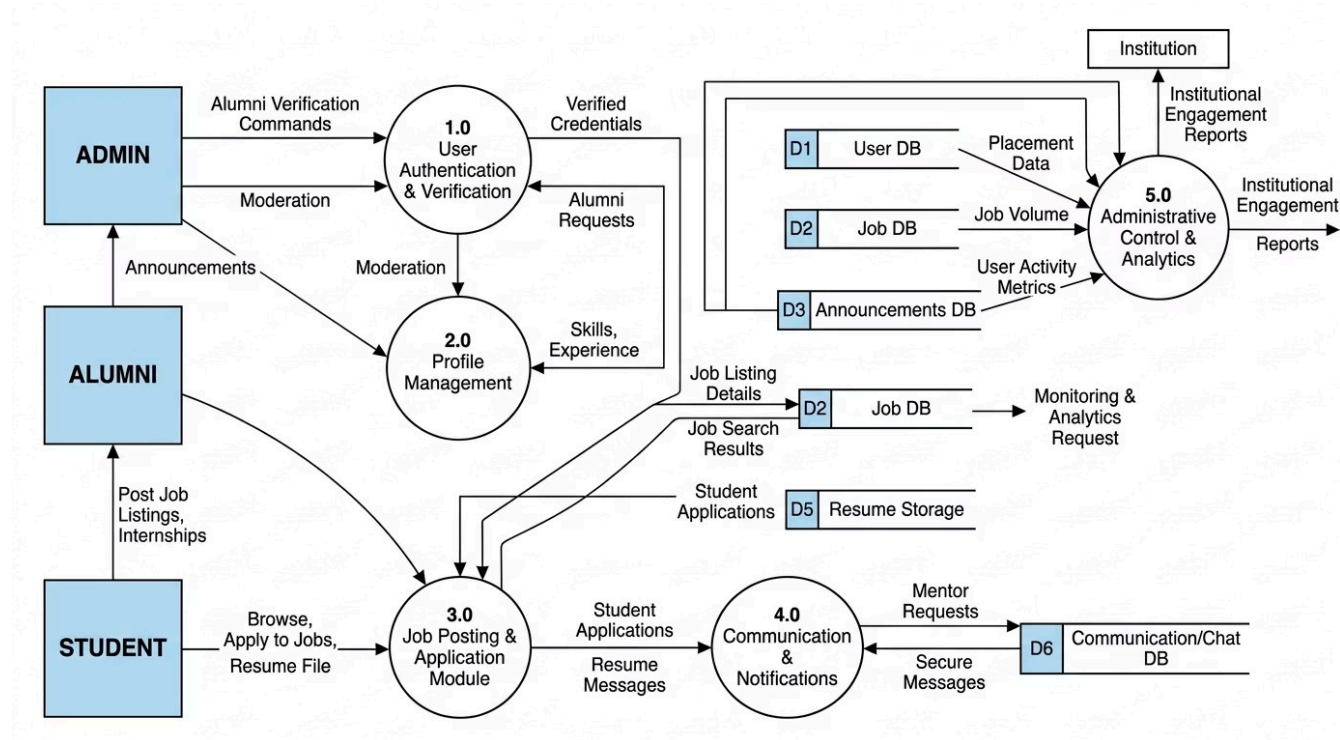
Profile & Admin Verification Module

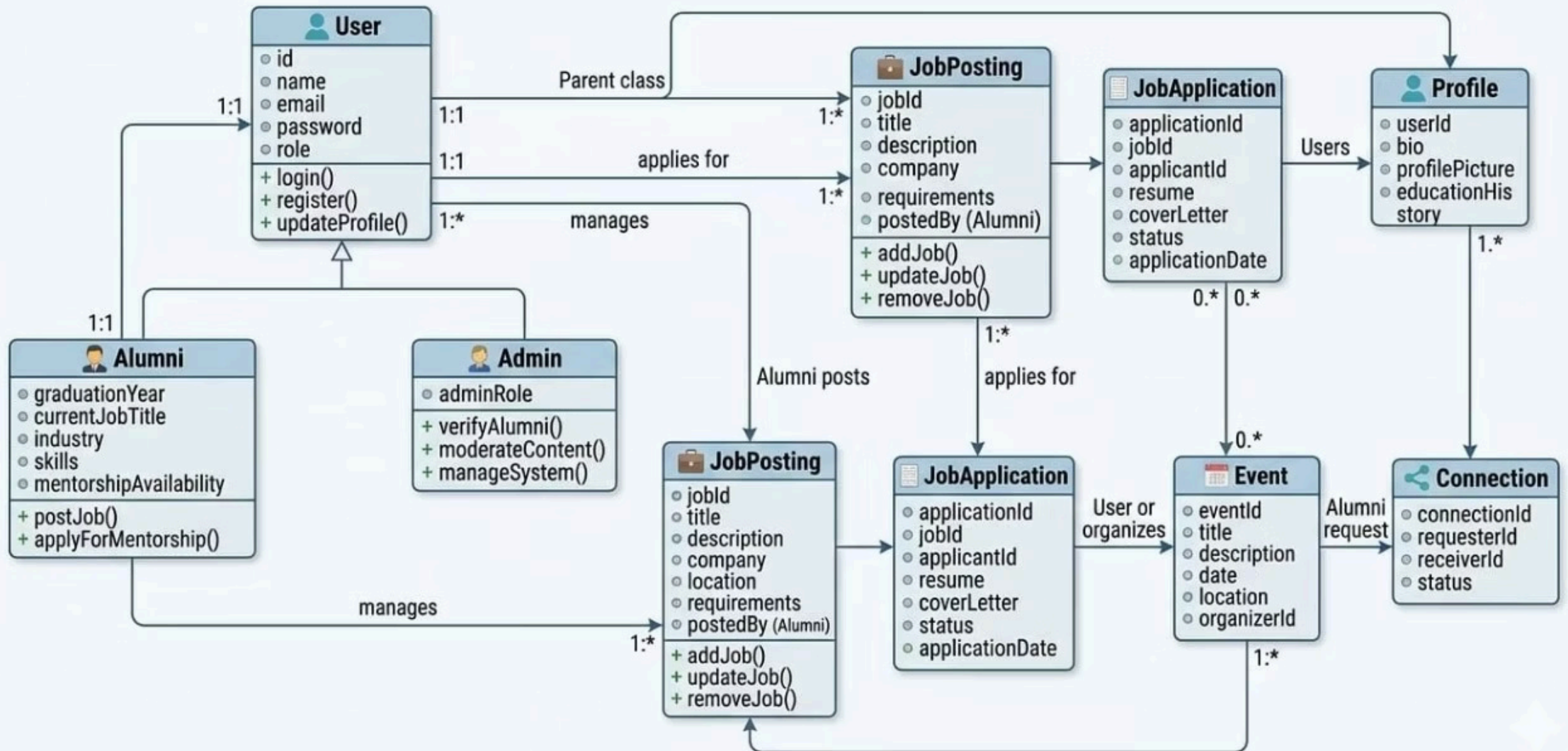
Ensures network authenticity. Alumni and students must be explicitly verified by the Admin before they can view directories or post jobs.

Secure Communication Module

Features a built-in one-on-one messaging system for direct mentorship, alongside a global announcement board for events.

Data Flow Diagram (DFD)

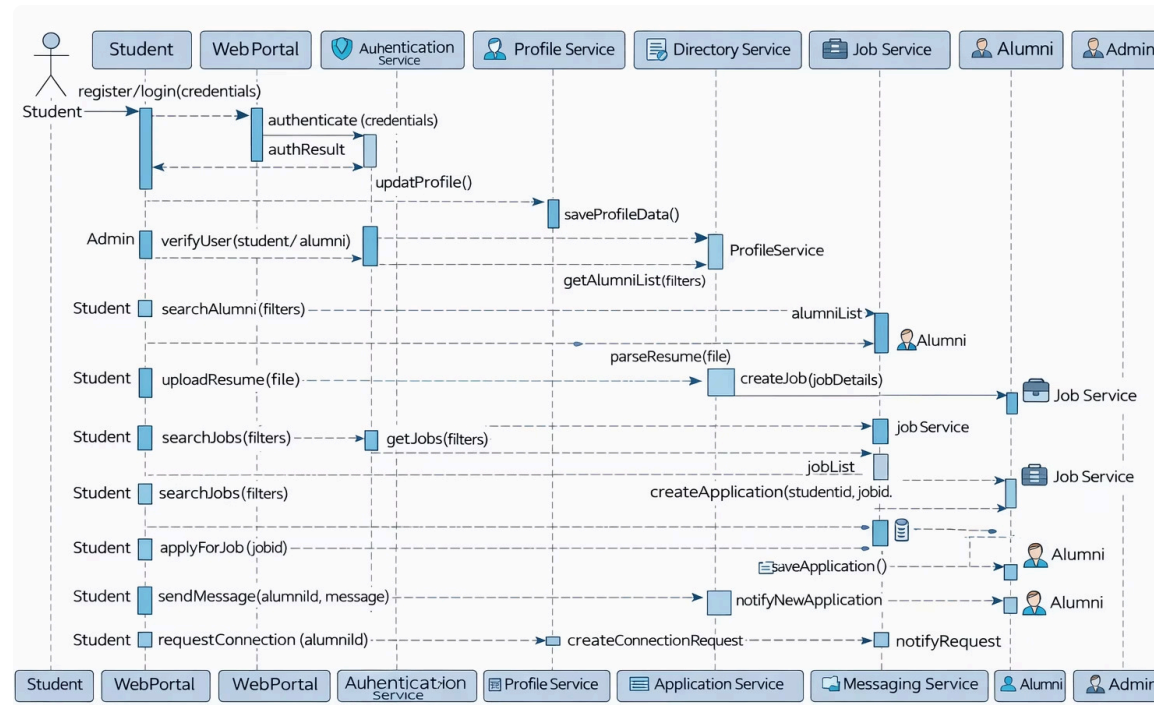




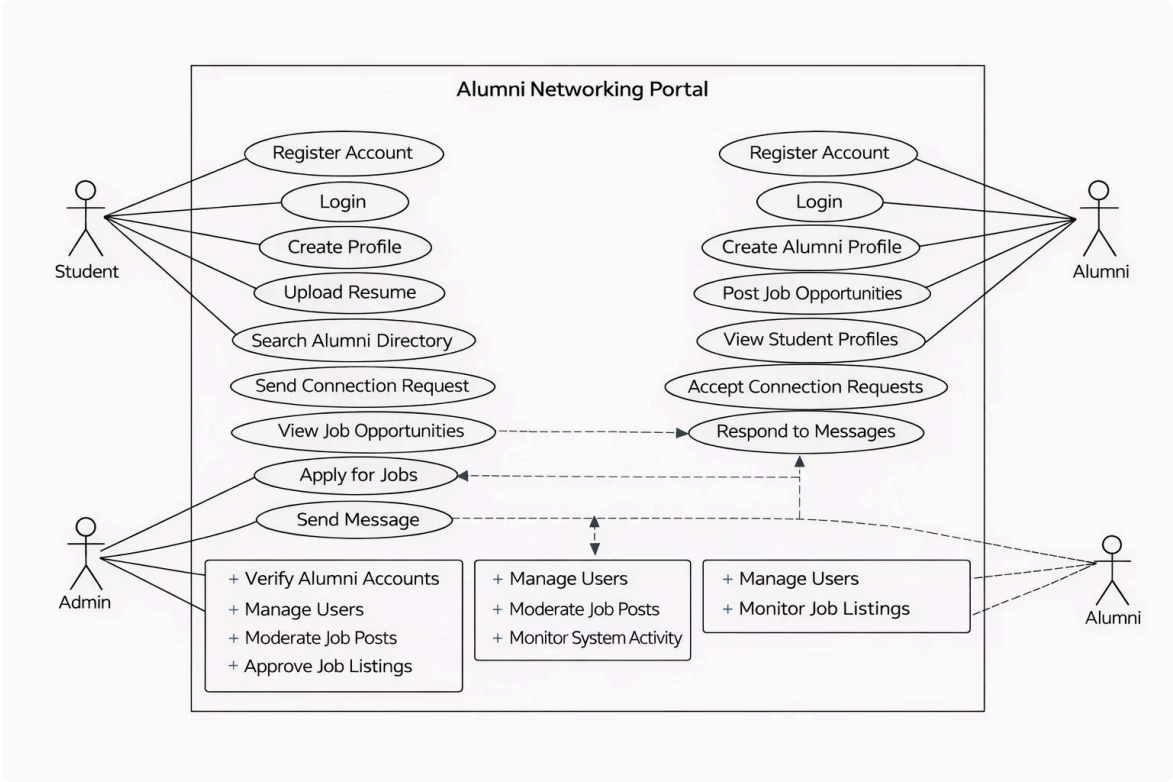
UML Diagrams

Class Diagram

3. Sequence Diagram



2.Usecase Diagram



Project Conclusion & Future Vision

The Alumni Networking Portal provides a secure and scalable platform for alumni-student engagement.

Project Summary

- Centralized, secure alumni networking platform
- Enhances job sharing and mentorship opportunities
- Features AI-based personalization for users
- Implemented a comprehensive role-based authentication system

Future Scope

- Development of a dedicated mobile application
- Integration of real-time chat & video mentoring
- Advanced AI analytics for deeper insights
- Scalability to support multiple institutions



Improves institutional placement support and alumni tracking



References

A compilation of academic literature and foundational texts that informed the development of the Alumni Networking Portal.

- 1 Alumni Networks: Fostering Engagement and Philanthropy**
Smith, J., & White, E.
Published: 2018
Explores strategies for building vibrant alumni communities and leveraging them for institutional support and career development.
- 2 Artificial Intelligence in Talent Acquisition: Reshaping the Future of HR**
Chen, D.
Published: 2020
Discusses the application of AI and machine learning in optimizing recruitment processes and candidate matching for organizations.
- 3 Digital Transformation in Higher Education: A Roadmap for the Future**
Jones, S., & Brown, M.
Published: 2021
Examines the role of technology in enhancing educational delivery, administrative efficiency, and overall student experience.
- 4 Deep Learning for Natural Language Processing**
Goldberg, Y.
Published: 2017
A foundational text covering neural networks and their application in understanding and generating human language, including semantic embeddings.
- 5 Building Microservices: Designing Fine-Grained Systems**
Newman, S.
Published: 2015
A comprehensive guide to designing and implementing scalable, independently deployable services for modern web applications.
- 6 AI-Powered Career Counseling: Enhancing Student Guidance**
Lee, A.
Published: 2022
Investigates how artificial intelligence can personalize career advice, match skills to jobs, and predict future career trends effectively.



Thank You

Questions and Discussion.

Avanthi Institute of Engineering and Technology
Makavarapalem

References



AI in Career Guidance: A New Era of Recruitment

Author: Doe, J., & Smith, A. (2023)

Category: AI & Recruitment



The Role of Alumni Networks in Higher Education

Author: Brown, L. (2022)

Category: Higher Education



Enhancing Career Success with AI Resume Review

Author: OpenAI Documentation (2024)

Category: AI & Career Development

Literature Review

A concise overview of key research and publications that inform our project's development, highlighting critical insights for alumni engagement and technology integration.

1

Designing Effective Alumni Networking Platforms

- **Author:** J. Smith, A. Chen
- **Year:** 2018
- **Key Point:** Emphasises the role of trust and reciprocity in fostering strong alumni networks for career development and social capital.

2

Leveraging AI for Personalized Job Recommendations

- **Author:** M. Johnson, L. Kim
- **Year:** 2020
- **Key Point:** Explores machine learning algorithms for matching job seekers with relevant opportunities based on skills, experience, and career trajectory.

3

Community Building in Online Professional Networks

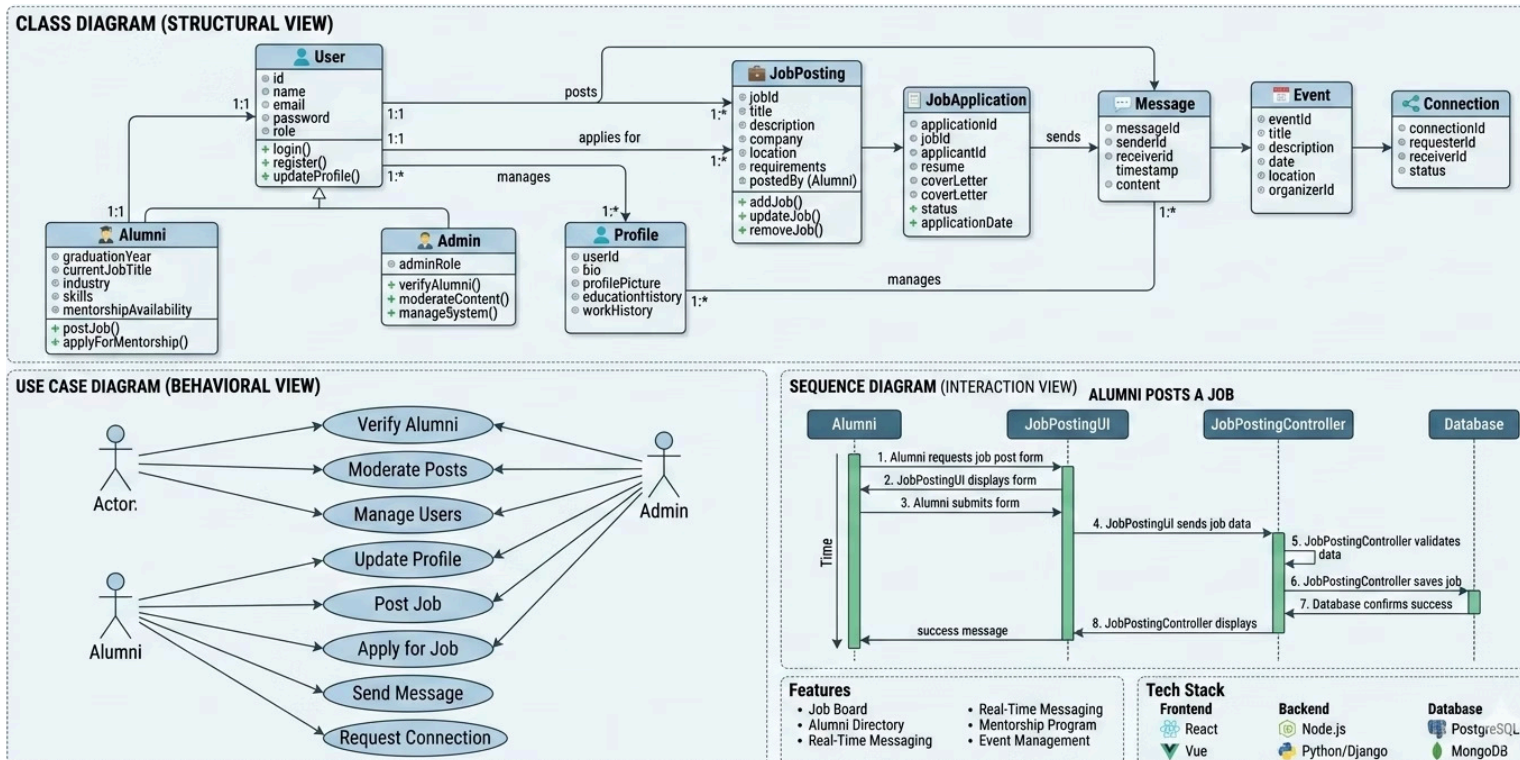
- **Author:** R. Davis, S. Gupta
- **Year:** 2019
- **Key Point:** Discusses strategies for enhancing user engagement and fostering a sense of belonging within digital professional communities.

4

Data Privacy and Security in Alumni Portals

- **Author:** T. Williams, B. Lee
- **Year:** 2021
- **Key Point:** Addresses critical considerations for protecting sensitive user data and ensuring compliance with privacy regulations in educational platforms.

UML Diagram: System Use Cases



PRESENTATION OVERVIEW

Presentation Flow

01

Introduction & Background

- Project Overview
- Key Objectives

03

Proposed System

- Solution Overview
- Core Features & Modules

05

Results & Impact

- Expected Advantages
- Anticipated Outcomes

02

Problem Analysis

- Existing System Deficiencies
- Identified Student Pain Points

04

System Design & Implementation

- Architecture Diagram
- Working Principle
- Technology Stack

06

Future Scope & Conclusion

- Enhancements
- Final Remarks

AI-Powered Features: Enhancing Alumni Connections

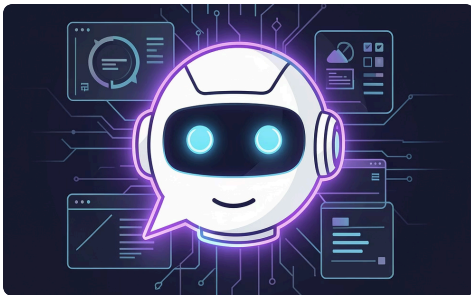
Our platform integrates cutting-edge AI features to create a dynamic and intelligent network, streamlining processes and fostering meaningful connections.



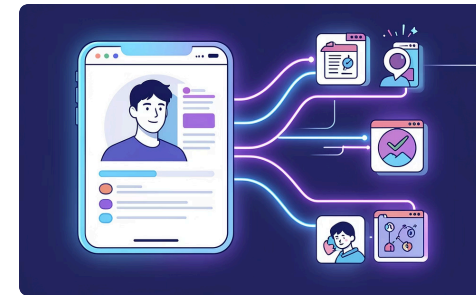
Smart Job Matching Based on Resumes



Alumni career trend analysis



Chatbot Support



Personalized Recommendations

How Our Solution Addresses Key Challenges



Verified Profiles

Administrative verification creates **trusted referrals** and ensures authenticity across the network



Centralised Portal

Single platform improves **placement visibility** and streamlines opportunity discovery



Direct Communication

Instant messaging enables **faster mentorship** connections and professional guidance



Data Analytics

Comprehensive insights support **institutional strategy** and career programme development



AI Matching

Intelligent algorithms deliver **personalised recommendations** and reduce manual effort

End-to-End System Workflow



FUTURE SCOPE

Future Scope & Innovation Ahead

-  — Advanced AI for personalised career guidance and mentorship recommendations.
-  — Mobile application expansion (Android/iOS) for wider accessibility.
-  — Integration with external platforms like LinkedIn and institutional ERP systems.
-  — Predictive analytics for career trend forecasting and placement insights.
-  — Scalability to support multiple institutions and departments.
-  — Enhanced admin dashboards for data-driven decision-making.
-  — Support for internships, research and alumni-funded initiatives.

Real-World Impact: From Alumni Data to Placement Outcomes

